Module 7 — Estimation & Confidence Intervals

Flipkart- 10M 2530

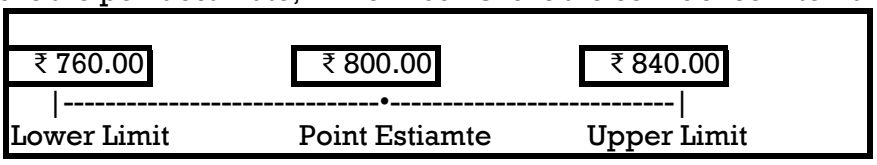
1000 F
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Business Scenario
Suppose 100 customers have an average order value of ₹800.

Can we confidently say that the AOV of all the customers is exactly ₹800?
No, ₹800, is our sample estimate, not necessarily the true population mean.

Point Estimate
A point estimate is a single numerical value calculated from sample data to estimate a population parameter.
Example:
Sample AOV = ₹800
Xbar=₹800

Interval Estimation
Interval estimation provides a range of plausible values for a populaiton parameter using the sample data nad a chosen conifence interval.
Example:
₹760≤μ≤₹840
Here, ₹800 is the point estimate, while ₹760-₹840 is the confidence interval.

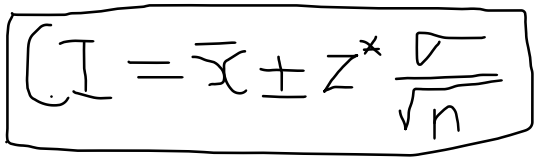


Which estimate communicates uncertainty better?
Interval estimation, because it shows a range rather than a single value.

Business Applications:
1. E-commerce: AOV
2. HR analytics: Average Emp Salary
3. Manufacturing: Average product weight.
4. Marketing: Campaign conversion rate.

Confidence Interval

A confidence interval is an interval calculated from sample that estimates a population at a specified confidence level.
For a population mean with known population standard deviation:



Xbar = Sample Mean
z* = Critical Z-Value
σ = Population SD
n = Sample Size

Example:
A company wants to estimate average delivery time.

Parameter	Value
Sample Mean	30 minutes
Population SD	10 minutes
Sample Size	100
Confidence level	95%

z* = 1.96

Step 1: Calculate the Standard Error

SE = σ/SQRT(n) =10/SQRT(100) 1

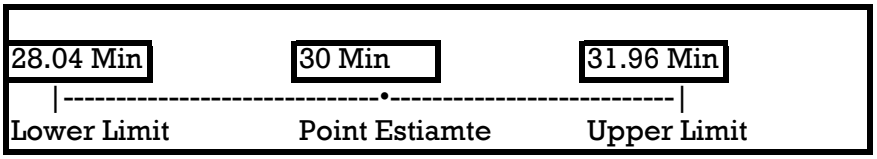
Step 2: Calculate Margin of Error

ME = 1.96 * SE =1.96*1 1.96

Step 3: Calculate Confidence Interval

CI = (30-1.96, 30+1.96)
(28.04,31.96)

95 % Confidence Interval



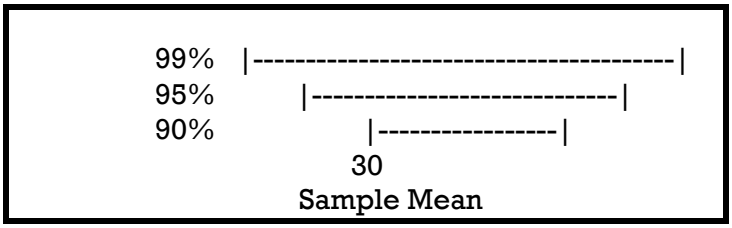
Interpretation:
The estimated population mean delivery time is between 28.04 and 31.96 minutes.

Confidence level
It is long run percentage of intervals that would contain the true population parameter if the sampling and interval- construction process were repeated manny times under required asumptions.

Confidence Level	Z-Score
90%	1.645
95%	1.96
99%	2.576

Sample Mean = 30 min SE =1

Confidence Level	Confidence Interval	
90%	28.355	31.645
95%	28.04	31.96
99%	27.424	32.576



90% : Narrower interval, lower confidence level
95% : Commonly used balance.
99% : Wider interval, higher level of confidence.

Margin of Error
It is the amount added and subtracted from a point estimate construct a confidence interval.

For mean when population SD is known:

$$ME = z * SE [\sigma/SQRT(n)]$$

For a proportion:

$$ME = z * \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

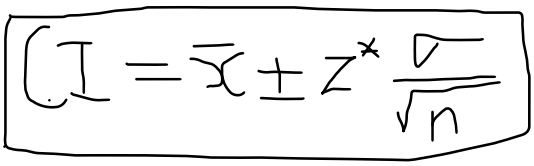
Supose:
z* = 1.96
σ=12
n=64
Then

SE = 12/SQRT(64) 1.5
ME = 1.96*1.5
ME = 2.94

Sample mean = 500

CI	
497.06	502.94

CI for Mean (SD/ σ Known)



Practical Case: Manufacturing

A factory produces bottles with target fill volume. We want to estimate the true average filling volume.

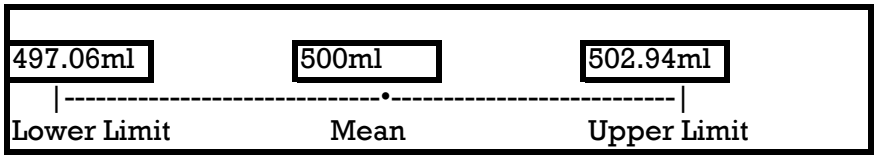
Parameter	Value
Sample Mean	500ml
Populaiton SD	12ml
Sample Size	64
Confidence level	95%

SE = 12/SQRT(64) 1.5

ME = 1.96*SE(1.5) 2.94

Confidence Interval
CI=500±2.94
497.06ml to 502.94ml

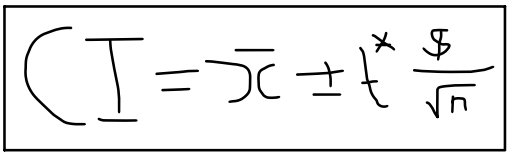
95 % Confidence Interval



Confidence Interval for Mean (SD unknown)

When the population sd is unknown, we generally estimate it usinng the sample standard deviation and use the t distribution to construct a confidence interval for the population mean.

s = Sample SD
t* = Critical t value
n-1 = degree of freedom
for one sample t-interval



Employees Salaries

Sample mean salary = 40000
Sample SD = 5000
Sample size = 25
Condidence interval = 95%

Degree of freedom = 24
Critical t value = Approximately 2.064

SE = 5000/SQRT(25) 1000

Margin of error = 2.064(1000) = 2064

CI = 40000+-2064
₹37,936 to ₹42064

Z interval		**T interval**		
SD known		unknown		
Uses Sd		Uses Sample SD		
Uses Z critical value		Uses T critical value		
no dependent		T value depends on Sample size		

Confidence Interval for Proportions

A confidence interval for a populatio proportion estimates a populaiton percentage, such as conversion rate, defect rate, using a sample proportion.

$$CI = \hat{p} \pm Z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

p = x/n
x is the number of successes and n is the sample size

Marketing Conversion Rate

An online campaign attracts 500 visitors . Of these 100 makes purchase.

p = 100/500 0.2
sample conversion rate is 20%

SE = SQRT(0.20(1-0.20)/500
0.017889

Margin of error
ME = 1.96(0.017889)
0.035062
Confidence interval 0.20+-0.035062
0.164938 0.235062
16.49% to 23.51%

Business Interpretation The interval estimates the underlying conversion rate, assuming the sampling process and normal approximation conditions are suitables.

Interpretation of Confidence Intervals

95% range --> 415-460
Sample1 Sample2 Sample3 …. Sample n
450 420 460 … 455

Interval A |----------•----------|
Interval B |----------•----------|
Interval C |----------•----------|

Which interval is widest?

C, the widest interval has the greatest width

Which interval gives the greatest precision?

B, the narrowest interval is more precise under comparable conditions